About the Project: AI Embryo Quality Assessment & Grading

Clinical Background, System Objectives & Architecture Guide for Web Development

1. Clinical Motivation & Project Goal

In In Vitro Fertilization (IVF), selecting the healthiest embryo for uterine transfer is the single most important factor determining pregnancy success. Currently, human embryologists inspect blastocysts under microscopes using manual grading systems (Gardner & Schoolcraft 1999; ALPHA/ESHRE 2011 Consensus). However, manual evaluation is highly subjective—inter-observer disagreement among senior embryologists is documented between 30% and 40%.

Project Objective: Build an objective, clinically validated Deep Learning System that automates blastocyst grading into three quality tiers (**Grade-1 Good**, **Grade-2 Fair**, **Grade-3 Poor**) and delivers immediate viability predictions (Transferable vs Non-Viable) with zero patient data leakage.

2. Clinical Quality Tiers (Gardner Standard)

The system categorizes human blastocysts based on developmental expansion, Inner Cell Mass (ICM) compaction, and Trophectoderm (TE) cellular junctions:

- **Grade-1 (Good Quality):** Gardner 3AA, 3AB, 4AA equivalent. Expanded blastocoel cavity, tightly packed distinct ICM, cohesive single-cell trophectoderm layer. Recommended for single blastocyst transfer (eSET) or priority freezing.
- **Grade-2 (Fair Quality):** Gardner 2BB, 3BB, 2AB equivalent. Moderate blastocoel expansion, loosely grouped ICM, fewer TE cells. Transferable candidate with good implantation potential.
- **Grade-3 (Poor Quality / Non-Viable):** Gardner 1BB, 1BC, degenerative embryos. Poor cellular cohesion, arrest, or severe fragmentation. Consensus recommends against transfer.

3. Dataset Overview & Strict Zero-Leakage Guarantee

• **Total Images:** 2,344 microscopic images across **851 patient cohorts**.
• **Patient-Separated Splitting:** All splits were partitioned strictly by *patient_id* using Stratified Group K-Fold. **No patient appears in both training and test sets.**
• **Splits:** Training (1,664 images / 607 patients) | Validation (355 images / 122 patients) | Held-Out Test (325 images / 122 patients).

4. Model Architecture: Heterogeneous Super-Ensemble

The project deploys a 4-way deep ensemble combining complementary neural representations:
1. **ConvNeXt-Tiny:** 7x7 depthwise convolutions capturing Inner Cell Mass compaction.
2. **DenseNet-121:** Dense connectivity preserving fine trophectoderm cell junctions.
3. **ResNet-50:** Deep 2048-dimensional residual bottleneck representations for blastocoel expansion.
4. **EfficientNet-B0:** Compound-scaled CNN with squeeze-and-excitation channel attention.
5. **5-View Test-Time Augmentation (TTA):** Invariant to rotation (0-360°) and illumination.

5. Website Developer Integration Guide

For your web application (Streamlit / Flask / FastAPI / React), the backend is packaged inside embryo_web_deployment/:
• **Pretrained Checkpoints:** In embryo_web_deployment/models/
• **One-Line Prediction:** Import EmbryoPredictor from inference_pipeline.py:

```
predictor = EmbryoPredictor()
result = predictor.predict('uploaded_image.png')
```

• **JSON Output:** Returns predicted_grade, confidence_percentage, clinical_viability, probabilities for all 3 grades, and clinical recommendations for display on the website UI.